

Getz CS-499 Milestone Four Narrative

Artifact Description

The artifact I am working on for database enhancement is the Company Data Dashboard for Grazioso Salvare. It is the same artifact used for the software design and engineering enhancement. The dashboard allows staff to easily filter through data from the Austin animal shelter to identify dogs that match certain criteria for their programs. It uses Dash for the front end and Flask for the backend, communicating via HTTP requests. I am using MongoDB Atlas to store my data. It was originally created in CS-340 Client/Server Development and was first completed on 2/17/26. The enhanced version was completed in CS-499 Computer Science Capstone on 7/16/26.

Artifact in ePortfolio

I wanted to include this artifact in my ePortfolio for a plethora of reasons, many of which were covered in the milestone two narrative. When I first made this application, it showed me how databases are built and how applications use them. Improving it would allow me to cover both the software design and engineering as well as the database categories in one artifact that I would enjoy working on. The database category is the most important for my career interests. Being able to take in real-world data, sort it, and display it in a working application is a useful skill in the data field. It shows end-to-end thinking as well as various data skills such as analytical thinking, data literacy, handling data flow, and managing databases (ETL) that I wanted to highlight. Many components I added showcase these skills, including the data.ipynb file, the Atlas database (Atlas user, etc.), and the CRUD file. MongoDB Atlas is one of the most important, as it is a tool (or similar) that I may use in the near future.

This enhancement builds upon what I did in the software design/engineering enhancement. I wiped my fake data from my Mongo database, loaded the Austin animal shelter API data into a pandas dataframe, cleaned it, and then loaded it into the Mongo database. I also made other small changes such as removing the Leaflet map and making the address links to Google Maps (see reflection section). Performing these changes makes the dashboard real and usable in the real world.

Course Outcomes

I believe that I did meet all the course outcomes that I had planned for this enhancement. I do not have any updates to my course outcomes at this time. I developed a security mindset by adding a few features. I first added a database user for the dashboard to apply the principle of least privilege. There is no reason the dashboard should have all admin rights to the database. I configured it in Atlas to make the user only have read/write or admin access for the animal's collection. I also used `quote_plus` to query Google Maps to help protect from API data being injected or the URL being changed. By using the `RESCUE_QUERIES` dict for the dog filters, NoSQL injections are prevented. I employed strategies for collaborative environments and to support decision-making by completing this application. And, I demonstrated the ability to use techniques, skills, and tools to deliver value and accomplish goals. I met these two outcomes by making my code modular and well documented, which allows future developers to build upon it. The CRUD layer lets a technical stakeholder change the rescue criteria in a central location. Now that the dashboard uses real data from the animal shelter, a real dog rescue company could use this dashboard. It reduces the time needed to sort through the shelter's database to find matches for their programs. The filters can easily be changed to match whatever any dog rescue company is looking for.

Reflection

To visually view the changes, I made in this enhancement visit my [GitHub repository commit page](#). Completing the work I had planned and started in the first enhancement was very enjoyable. I learned a lot from this enhancement and was able to leverage a lot of the skills I have built. I already had the database set up and connected to the application, but I had it loaded with fake data. The first thing I did was load the API data into a pandas dataframe to clean and process it. I did this for a few reasons; first, I originally had in mind to use aggregation pipelines to sort and filter the data but decided against this due to the size and formatting of the data. The animal shelter only had around 15,000 rows, some of which were duplicate animals. The API data also had a different schema than what was originally used for the dashboard, so I had to make some changes. And the Dash application uses a dataframe, so it allowed me to easily view how the data will look.

I originally wanted to add an index to the database too but decided it was also unnecessary. The most time-consuming part of loading the dashboard is the initial load with no filter. It loads all 15k entries into the Dash dataframe. For the scope of the application, this is fine because the load speed is hardly noticeable, and the filters load significantly faster once the data is loaded into the dataframe. To improve the speed of the dashboard, we could add an index for the dog filters, sort and filter the data before loading it into the database so unnecessary data wouldn't be added in the first place, or we could configure the API and CRUD module for performance.

One of the first things I noticed was that the API data did not have coordinates and only had an address. This presented an issue because the Leaflet map relied on coordinate data. I tried to convert the addresses into coordinates but quickly learned that much of the data has

incomplete addresses or none at all. A simpler and more efficient solution I came up with was to add a link to Google Maps. Instead of a prebuilt map, a user can simply click on the address and load Google Maps in a new tab. The incomplete addresses can still be loaded, and Google Maps will show a general area, further improving usability. Users can save addresses, look in the surrounding area, or get directions easier. There were many similar data formatting issues that I had to take care of. The API data had many columns that were redundant. To make the dashboard easy to read, I combined many of these columns. There were also other columns such as `age_in_weeks` and `sex_upon_outcome` that I needed to create. Many of the columns had null values such as the name, sex, or age columns. To improve the user experience, I filled these values with strings such as “No name” or “unknown” instead of null or nan. Due to my experience in data analytics, formatting and filtering the data in a way the dashboard could use was time-consuming but straightforward. I soon realized that mixing date and string data worked for formatting but was not working well with the rest of the application. I made some changes to store the dates as BSON dates and moved converting nulls to strings in the Dash application. I also added a `$jsonSchema` validator to ensure all fields are the correct datatype.

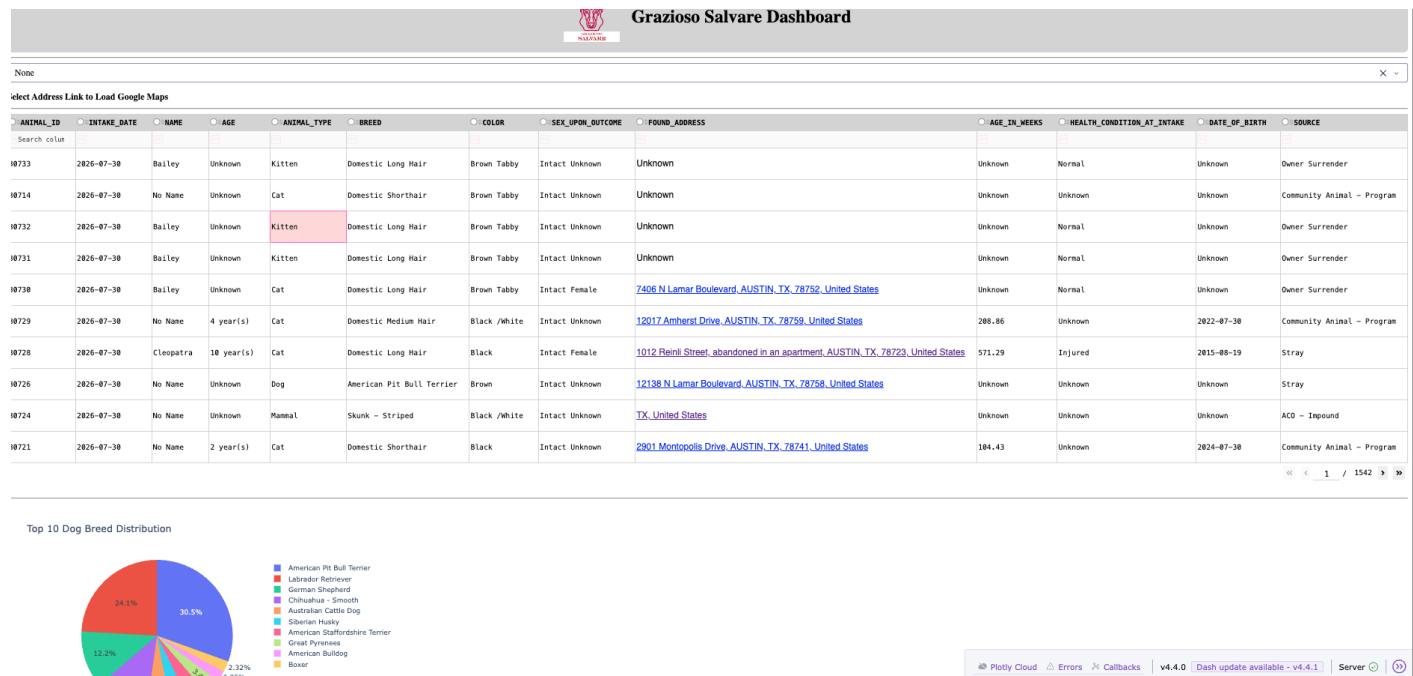
After formatting and cleaning the data, I ran into a problem. The way my CRUD module was set up, I had no way to load all the data at once. I had to create a new method in my CRUD module named “`load_database`” that uses PyMongo’s `insert_many` function. I set it up so that when the data is loaded, the database is first deleted to prevent duplicates. The deletion and loading of the database also has no safety net. If the API fails, the database will be empty because I did not create a backup. For the scope of this application, this is fine. The data is not being loaded automatically. To make the dashboard even more robust, we could add a way for

the data to be loaded weekly or even daily. We could also add a safety net for deleting and loading the database.

The original database user for the application was set to read-only, which prevented me from loading the data. I had to go to Atlas to configure the user to be read and write so I could load the data from this application. To apply the JSON schema, I configured the user to have that specific privilege as well. With that method and the database user in place, I was able to load the API data into my Mongo database without any problems. I then made various changes to the dashboard such as removing the Leaflet map, removing the radio buttons, and reordering the columns. I also ordered the animals by intake date so the newest animals will be shown first.

Screenshots

Full zoomed out dashboard with no filter



MongoDB Atlas homepage showing creation of aac database and animals collection,

ORGANIZATION
Creston's Org - 2026-...

PROJECT
Project 0

animals

oac

+

Cluster0 > oac

View monitoring

Visualize your data

Create collection

Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
animals	-	2.91 MB	6.02 MB	15K	390.00 B	1	2.92 MB

CLUSTERS (1)

Search clusters

Cluster0

oac

animals

admin

local

ORGANIZATION
Creston's Org - 2026-...

PROJECT
Project 0

animals

+

Cluster0 > oac > animals

View monitoring

Visualize Your Data

Documents

15K

Aggregations

Schema

Indexes

Validation

Type a query: { field: 'value' } or [Generate query](#)

Explain

Reset

Find

Options

ADD DATA

BULK

EXPORT CODE

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```
_id: ObjectId('6a6c960d323a0b0812931b06')
intake_date: 2025-06-16T11:16:00.000+00:00
animal_id: "10000"
animal_type: "Cat"
source: "Community Animal - Program"
sex_upon_outcome: "Intact Male"
breed: "Domestic Shorthair"
date_of_birth: 2023-06-16T08:00:00.000+00:00
found_address: "4818 E Yager Lane Apt 5111, MANOR, TX, 78653, United States"
name: "No Name"
health_condition_at_intake: "Unknown"
age_in_weeks: 163
color: "Brown Tabby /White"
age: "3 year(s)"
```

```
_id: ObjectId('6a6c960d323a0b0812931b07')
intake_date: 2025-06-16T11:20:00.000+00:00
animal_id: "10003"
animal_type: "Cat"
source: "Community Animal - Program"
sex_upon_outcome: "Intact Female"
breed: "Domestic Shorthair"
date_of_birth: 2023-06-16T08:00:00.000+00:00
found_address: "14514 Highsmith Street, AUSTIN, TX, 78725, United States"
name: "No Name"
health_condition_at_intake: "Unknown"
age_in_weeks: 163
color: "Grey"
```

Alert System Status: All Good

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Screenshot of water rescue filter on dashboard.



Grazioso Salvare Dashboard

Water Rescue

Select Address Link to Load Google Maps

ANIMAL_ID	INTAKE_DATE	NAME	AGE	ANIMAL_TYPE	BREED	COLOR	SEX_UPON_OUTCOME	FOUND_ADDRESS	AGE_IN_WEEKS	HEALTH
29392	2026-07-20	Yumi	1 year(s)	Dog	Labrador Retriever	Yellow	Intact Female	AUSTIN, TX, 78702, United States	56	Normal
30895	2026-07-16	Bradi	2 year(s)	Dog	Labrador Retriever	Black /Tan	Intact Female	2300 Rosewood Avenue, AUSTIN, TX, 78702, United States	146.57	Unknown
29538	2026-07-06	Barbie	1 year(s)	Dog	Labrador Retriever	White /Cream	Intact Female	6201 Kellam Road, DEL VALLE, TX, 78617, United States	55.71	Unknown
29392	2026-07-04	Yumi	1 year(s)	Dog	Labrador Retriever	Yellow	Intact Female	7201 Levander Loop Apt a, truck port, AUSTIN, TX, 78702, United States	56	Normal
27877	2026-07-01	Honeybear	1 year(s)	Dog	Labrador Retriever	Yellow Gold /Cream	Intact Female	AUSTIN, TX, 78745, United States	60.71	Normal
27143	2026-06-02	Malibu	2 year(s)	Dog	Labrador Retriever	Black /White	Intact Female	129 Mark Adams Road, DEL VALLE, TX, 78617, United States	112.57	Unknown
27126	2026-06-02	Sugar	12 months	Dog	Labrador Retriever	Cream /Tan	Intact Female	Pandora Street, sol wilson 12th and oak springs, AUSTIN, TX, 78702, United States	51.86	Unknown
27077	2026-06-01	Honeybear	1 year(s)	Dog	Labrador Retriever	Yellow Gold /Cream	Intact Female	8700 Brodie Lane, AUSTIN, TX, 78745, United States	60.71	Unknown
26769	2026-05-28	Gancito	8 months	Dog	Labrador Retriever	Brown	Intact Female	12505 Briarcreek Loop, MANOR, TX, 78653, United States	34.71	Normal
21481	2026-05-26	Cookie	7 months	Dog	Labrador Retriever	Black /White	Intact Female	1401 E Rundberg Lane, AUSTIN, TX, 78753, United States	31.14	Normal

Plotly CloudErrorsCallbacks v4.4.0 Dash update available - v4.4.1 Server

Screenshot of mountain or wilderness rescue pie chart and bottom of dateable.

29829	2026-07-11	No Name	2 year(s)	Dog	German Shepherd	Brown /Black	Intact Male	17817 Powder Creek Drive, MANOR, TX, 78653, United States	107.14	Normal
29827	2026-07-11	Ranger	8 months	Dog	Siberian Husky	Cream /White	Intact Male	5701 Johnny Morris Road, AUSTIN, TX, 78724, United States	37.29	Normal
29293	2026-07-02	Copper	11 months	Dog	German Shepherd	Black /Brown	Intact Male	TX, United States	47.43	Unknown
29292	2026-07-02	Kahu	7 months	Dog	German Shepherd	Black /Tan	Intact Male	TX, United States	30	Unknown
29249	2026-07-01	No Name	1 year(s)	Dog	German Shepherd	Sable Fawn/Sooty	Intact Male	E Meadow Bend Dr & View Ridge Dr, AUSTIN, TX, 78724, United States	82.29	Unknown
28795	2026-06-24	Lingvine	1 year(s)	Dog	German Shepherd	Black /Tri Color	Intact Male	2300 W Ben White Boulevard, AUSTIN, TX, 78704, United States	57.43	Unknown
28717	2026-06-23	Willy	7 months	Dog	German Shepherd	White	Intact Male	Cameron Road, AUSTIN, TX, 78723, United States	31.43	Unknown
28677	2026-06-22	Spanky	2 year(s)	Dog	Siberian Husky	White /Ghost Grey	Intact Male	6625 McKinney Falls Parkway, AUSTIN, TX, 78744, United States	109.86	Unknown

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